PEAK_TRADE

Unified Runtime, Market Intelligence, Meta-Learning & Optimization Blueprint

Public Data Plane + Private State Plane + Market Intelligence + Learning + Optimization + Meta-Learning + Governed Productive Parameter Lineage

Unified blueprint date: 2026-09-26 | Source documents: Runtime Blueprint (Track D Updated) + Meta Learning & Optimization Universe Concept v3.3 Expanded

AUTHORITY=NONE
This document consolidates planning and implementation sequence only. It does not supersede the Master Runbook, does not create a new SSOT, and does not itself authorize trading, selection, promotion, deployment, risk override, or external effects.

Canonical end-state principle

Peak_Trade should learn the market and learn from its own research without turning Learning or Optimization into a second trading engine. Market Intelligence produces calibrated evidence. Learning turns realized outcomes into reusable evidence. Optimization tests bounded hypotheses. Meta-Learning improves future research choices. Governance and Risk control productive admission. Master V2 + Double Play remain the sole trading-decision authority.

Boundary	Unified position
Market / Private Data	Facts and state only.
Market Intelligence	Forecast/evidence only; forecast is not decision.
Learning	Outcome/evidence learning; no trading or promotion authority.
Optimization	Research, candidates, robustness and proposals; no direct productive write.
Meta-Learning	Improves research/search choice; no trading authority.
Governance + Risk	Admission/authorization boundary for explicitly permitted productive parameters.
Master V2 + Double Play	Sole trading-decision authority.
Execution	External effects only through existing explicit wire boundary.

LIVE_ENABLED=false | LIVE_ARMED=false | PRIVATE_WS_ORDER_SEND_AUTHORIZED=false | MULTI_FUTURE_RUNTIME_AUTHORIZED=false | MAX_POSITIONS_EFFECTIVE=1

1. Why the two blueprints are one architecture

The Runtime Blueprint defines the trustworthy data/state substrate and the Market-Intelligence direction. The Meta-Learning & Optimization Concept defines how evidence is learned, researched, returned, governed, and - only after explicit authorization - allowed to reach a real productive parameter consumer. They are not competing plans; they are successive layers of one dependency graph.

Dependency law: trustworthy facts/state first -> Market Intelligence second -> realized outcomes and Learning third -> Optimization/Meta-Learning fourth -> Governance/Authorization fifth -> productive parameter consumption last.

1.1 Unified architecture

```
OKX EEA
|
+-- PUBLIC DATA PLANE --------> canonical market facts/history
| |
| v
| MARKET INTELLIGENCE
| state / forecast / uncertainty
| |
| v
| REALIZED N_BARS OUTCOME
| |
| +------------+-------------+
| | |
| v v
| LEARNING LOOP OPTIMIZATION LOOP
| evidence/calibration experiments/robustness
| ^ |
| | v
| +------ META-LEARNING <----+
| |
| v
| Candidate + Evidence + PROPOSAL_ONLY
| |
| v
| GOVERNANCE + RISK
| | explicit authorization
| v
| GOVERNED PRODUCTIVE PARAMETER
| | typed authorized seam
| v
| CURRENT PRODUCTIVE CONSUMER
|
+-- PRIVATE STATE PLANE ------> account/position/order/fill state

MASTER V2 + DOUBLE PLAY = sole Trading-Decision Authority
EXECUTION = only existing governed external-effect boundary
```

1.2 Non-negotiable separation

• UNIVERSE != SURFACE: the Optimization Universe is the research container; each optimizable surface requires explicit bounded authorization.

• EXPERIMENT_SELECTION != TRADING_SELECTION: research chooses experiments, Cap 2.3 chooses the productive future.

• CANDIDATE != PRODUCTIVE_CONFIGURATION: a winning candidate remains evidence until separately authorized.

• META_EVIDENCE != AUTHORITY: Meta-Learning may improve the next experiment/search choice, never the trade decision.

• Private authenticated connectivity != send authority; Public/Private planes do not create decision authority.

2. Foundation - Public and Private Runtime Planes

The first dependency is trustworthy, replayable, provenance-rich facts and state. Market Intelligence, Learning and Optimization must consume canonical history/readmodels rather than transport sessions directly.

2.1 Public Data Plane

Capability	Unified requirement
Canonical facts	Instrument identity, timestamps/provenance, mark, last trade/trades, best bid/ask, finalized PT1M mark, OHLCV, data quality.
Transport	Public WS for continuous facts; REST for bootstrap, history, deterministic gap-fill and recovery.
Durability	Finalized PT1M/history store and historical query surface.
Semantic laws	Mark != last != generic candle close; finalized != in-progress; missing intervals explicit; no silent fill.
Consumers	Landscape, Ranking/B05/Cap2.2, O4/N_BARS/Learning, Research/Backtest, Optimizer through adapters.

2.2 Private State Plane

Capability	Unified requirement
Normalized state	Account/config, balance/equity observation, position, order, fill, quality, reconciliation, provenance.
Lifecycle	READ credential -> REST baseline -> persist/reconcile -> observation-only Private WS -> deltas -> REST recovery on gap/disconnect.
Security	Least-privilege read credentials; no secret values in facts/logs; sign capability != send authority.
Authority	No Private WS order send. Existing POST/wire permits unchanged.

Observation scope may cover multiple governed futures without authorizing multi-future trading. Cap 2.3 remains the sole productive selection owner.

2.3 Runtime convergence

- Consumers must not retain competing transport-specific truths where canonical adapters exist.
- Restart/replay/reconciliation must preserve deterministic semantics.
- Ranking, MV2/DP, Pretrade, Execution and live boundaries remain unchanged.

3. Track D1 - Market Intelligence

The primary subject is market behavior, not rewriting the trading strategy. Market Intelligence should answer what is happening now, what behaviors are plausible next, how uncertain that statement is, how well similar forecasts were calibrated historically, and when evidence is insufficient or drifting.

Capability	Target semantics	Authority
Market State	Trend, volatility, regime/transition, feature state, liquidity/spread where data supports them.	Observation only
Market Behavior	Forward-return distribution, continuation/reversal/range behavior, MFE/MAE, realized volatility, tails, transitions.	Evidence only
Probabilistic Outlook	Multi-horizon conditional distributions/probabilities, not a single deterministic price target.	Forecast only
Uncertainty / Coverage	Support count, effective coverage, uncertainty, evidence age, insufficient-evidence disposition.	No gating authority
Calibration	Forecast vs realized outcome, forecast error, reliability/calibration state.	Learning evidence
Drift	Distribution shift, state-performance drift, out-of-distribution evidence.	Observation/evidence

3.1 N_BARS as realized-outcome backbone

STATE(t) -> FORECAST(N) -> REALIZED_OUTCOME(t+N) -> ERROR/CALIBRATION -> LEARNING EVIDENCE

N_BARS remains the normative realized-outcome horizon mechanism. Multi-horizon expansion must preserve explicit horizon identity, PT1H semantics where ratified, finalization, overlap control and lookahead prevention.

3.2 Abstention is a valid result

A model must be allowed to return insufficient evidence when support is weak, calibration unhealthy, distribution shift material, or model disagreement excessive. Prediction quality is prediction + uncertainty + calibration + coverage, not confidence alone.

3.3 Decision Attribution is subordinate

Existing selected-future, Bull/Bear state, SideState/transition state and configuration references may be composed into attribution evidence to measure when Market Intelligence was useful. Ownership is referenced, not duplicated: COMPOSE_REFERENCES_DONT_DUPLICATE_OWNERSHIP.

4. Track D2 - Learning and the two Evidence Loops

Implementation must reuse CURRENT loop topology rather than build a parallel greenfield stack. The combined blueprint treats the two established loops as the core learning mechanism.

4.1 Loop A - Productive Learning / Outcome Loop

PRODUCTIVE OBSERVATION/DECISION -> OUTCOME / SYSTEM EVIDENCE -> SELF-LEARNING -> LEARNING EVIDENCE -> OPTIMIZATION INPUT

This loop learns from realized market/system outcomes. In the Market-Intelligence extension, forecast state and forecast evidence are joined with realized N_BARS outcomes to produce calibration, reliability, drift and conditioned behavior evidence.

4.2 Loop B - Meta-Learning / Research Loop

OPTIMIZATION EXPERIMENT -> CHALLENGER / OOS / ROBUSTNESS EVIDENCE -> META-LEARNING -> META-EVIDENCE -> BETTER NEXT RESEARCH/SEARCH CHOICE -> ...

This loop learns from the optimization process itself: which search families, surfaces, evidence patterns and experiment choices tend to produce robust or repeatedly rejected results. Its output may prioritize bounded research; it does not choose trades.

4.3 Inter-loop joins

Edge	Required classification
Learning Evidence -> Optimization Input	Typed evidence boundary; preserve provenance and identity.
Optimization Evidence -> Meta-Learning	Experiment/search/robustness/failure evidence.
Meta-Evidence -> Optimization	Bounded next experiment/search choice only.
Optimization Result -> Productive Stack	PROPOSAL_ONLY until Governance/Risk + explicit authorization.
Market Intelligence -> Learning	Forecast/evaluation evidence; no decision authority.
Learning/Optimization -> MV2/DP	No direct mutation; only separately governed supporting input/parameter seams.

Evidence feedback is not promotion. Research experiment selection is not trading selection.

5. Track D3 - Optimization Universe

There is one Optimization Universe containing multiple isolated surfaces. A code path, historical experiment, or universe membership does not itself authorize a surface.

5.1 Surface model

Surface / family	Unified status rule
F1 / M9 - Volatility Numeric Max-Age	Established vertical proof family; real productive consumer lineage is the reference pattern.
F2 - Fee/Slippage Cost Grid	Research candidate; materialize only under explicit surface ratification and proven consumer relevance.
F5 - Pure-Stack Numeric Calibration	Deferred/conditional; decompose and adjudicate safety/survival numerics separately.
F3 - Strategy Hyperparameter Research	Deferred/conditional; requires authority-boundary adjudication before touching protected trading semantics.
OLS coefficients	Diagnostic/evidence support; not an optimization surface.
Funding	Evaluation/model input; no proven optimization axis by existence alone.
Risk / Sizing	Constitutional boundary; NOT OPTIMIZABLE under this blueprint.
Legacy/unsupported search utilities	Preserve as historical/research evidence only; do not activate implicitly.

5.2 What Optimization should optimize first

• Features and state representations that robustly predict future market behavior out of sample.

• Forecast horizons and model families with stable calibration and coverage.

• Research parameters that have explicit bounded domains and a real or explicitly planned consumer.

• Robustness and abstention behavior before any attempt to maximize trading PnL.

Optimization produces candidates and evidence. It does not write productive configuration, authorize itself, select instruments, mutate MV2/DP, override Risk, or send orders.

6. Forecast Quality, Experiment Evidence and Anti-Overfit

Market Intelligence and Optimization are only useful if their claims survive temporal and statistical validation. The same evidence discipline must cover both forecasting and parameter research.

Control	Required role
Point-in-time separation	Prevent future information entering features, labels, state or candidate evaluation.
Walk-forward	Measure stability through time rather than one train/test split.
Overlap control	Purging/embargo or equivalent where outcome horizons overlap.
Multiple-testing awareness	Record full search space and trial count, not only the winner.
Challenger / OOS	Separate discovery from evidence used for acceptance.
Monte Carlo / Stress	Test path, sampling and adverse-condition sensitivity.
Cost sensitivity	Prevent apparent robustness that disappears under plausible fees/slippage.
Regime/state stability	Measure whether evidence is conditional or broadly stable.
Calibration	Probability forecasts must match realized frequencies within evidence limits.
Coverage / abstention	Unsupported intelligence must be allowed to abstain.
Drift	Detect mismatch between current observations and evidence population.

6.1 Experiment and Failure Memory

Rejected hypotheses and failed experiments are first-class evidence. Persist search space, trial count, failure reasons, walk-forward/stress/robustness failures and materially new evidence that justified a retry. Meta-Learning should reduce repeated rediscovery of already rejected hypotheses.

6.2 Counterfactual / shadow research

Historical replay may compare observed outcomes with alternative allowed model or parameter hypotheses. This remains offline research with COUNTERFACTUAL_AUTHORITY=NONE and cannot create a direct runtime decision path.

7. Productive Parameter Lineage - the End Goal

The Optimization Universe is not intended to remain an isolated research playground. Its long-term productive relevance is to improve explicitly authorized parameters that have a proven real consumer, while preserving strict authority separation.

7.1 Complete lineage

```
OPTIMIZATION SURFACE
-> Candidate Parameter Value
-> Search / Challenger / OOS / Robustness / Economic / Failure Evidence
-> PROPOSAL_ONLY
-> Governance + Risk Admission
-> Explicit Productive Authorization
-> Governed Productive Parameter / Configuration
-> Authorized Productive Parameter Seam
-> concrete CURRENT Productive Consumer
```

A surface is end-to-end closed for the productive target only when this complete typed lineage is auditable and replayable. Good performance alone is not productive relevance.

7.2 Governance and Risk boundary

Stage	Rule
Optimization Surface	Only explicitly bounded/authorized research domain.
Candidate + Evidence	Research/proposal only.
Governance + Risk	Checks evidence, constraints, target binding and hard risk boundaries.
Explicit Authorization	Binds candidate identity, evidence lineage, target, constraints and authorization identity.
Governed Productive Configuration	May arise only from valid authorization; no Optimization direct write.
Productive Consumer	Consumes the authorized parameter through an existing or explicitly ratified typed seam.
MV2 + Double Play	May consume permitted governed inputs while retaining sole trading-decision authority.

7.3 F1/M9 as reference vertical proof

Volatility Numeric Max-Age is the reference pattern for a real surface-to-consumer lineage: candidate max-age -> evidence -> governance/authorization -> governed parameter -> authorized seam -> actual volatility-presence consumer. It proves the shape of the end-state; it does not mean all other surfaces are complete.

Risk/Sizing remains constitutional and NOT OPTIMIZABLE here. Any future change to that status is a separate authority decision, not an implied continuation of Track D.

8. Unified Authority Matrix

Surface	May observe / learn	May decide/select	May cause external effect
Public Data Runtime	Public venue facts across governed observation scope.	No.	No.
Private State Runtime	Authenticated account/config/balance/position/order/fill state.	No strategy/selection decision.	Observation/reconciliation only.
Market Intelligence	Facts, state, forecasts, realized outcomes, calibration/drift.	No trading decision.	No.
Self-Learning	Outcome/system/forecast evidence; internal bounded learning state.	No trading/selection/promotion.	No.
Optimization	Experiments, candidates, challengers, robustness, proposals.	Research experiment choice only.	No direct productive effect.
Meta-Learning	Experiment/failure evidence and research-choice quality.	Bounded research prioritization only.	No.
Governance + Risk	Evidence, constraints, target lineage.	May explicitly authorize permitted productive configuration within its defined authority.	No order-send by itself.
Cap 2.3	Ranked candidates + existing inputs.	Sole productive selection owner.	No direct wire effect.
Master V2 + Double Play	Governed trading inputs/supporting intelligence if separately bound.	Sole trading-decision authority.	No direct wire effect.
Pretrade / Execution	Existing governed state inputs.	Existing authority only.	Only through explicit wire boundary and authorization gates.

OPTIMIZATION_PROMOTION_AUTHORITY=NONE | OPTIMIZATION_TRADING_DECISION_AUTHORITY=NONE | OPTIMIZATION_DIRECT_PRODUCTIVE_WRITE=FORBIDDEN | NO_SELF_DEPLOY=true | RISK_IS_HARD_BOUNDARY=true

9. Canonical Abarbeitungsreihenfolge

This sequence merges the Runtime/Track-D plan with the Meta-Learning/Optimization roadmap. Earlier completed work is retained as dependency evidence; CURRENT status must always be re-proven against origin/main before execution.

Phase	Action	Exit / stop condition
0 - Baseline	Validate CURRENT origin/main, Master Runbook, authority pins, existing owners.	No drift; no authority conflict.
1 - Public substrate	Complete/retain Public Data Plane, durable history and historical query adapters.	No competing market truth; deterministic recovery/replay.
2 - Private substrate	Complete/retain Private State observation/reconciliation plane.	No WS send; POST/wire gates unchanged.
3 - Convergence	Close consumer handoffs across public/private planes.	No duplicate runtime truth; authority unchanged.
4 - D01 Census	Prove CURRENT Learning, N_BARS, Evaluation, Optimization and promotion boundaries.	CURRENT topology known.
5 - D02 Closed-loop adjudication	Reconstruct Learning loop, Optimization/meta loop, every inter-loop join, identity/state/persistence/iteration semantics.	All CURRENT loops/joins classified; no invented owner.
6 - D03 Market Intelligence contracts	Ratify minimal Market State / Behavior / Forecast / Uncertainty / Calibration / Drift contracts only where CURRENT supports them.	No duplicate SSOT; N_BARS semantics preserved.
7 - Market Intelligence runtime	Implement largest coherent observation/research package and forecast-to-realized-outcome evaluation.	Forecast is not decision; abstention works.
8 - Learning integration	Feed calibrated forecast/outcome evidence into existing Learning evidence/export path.	No trading/promotion authority.
9 - Optimization integration	Use Learning input for bounded feature/state/horizon/model/surface research.	Proposal only; no productive write.
10 - Meta-return loop	Return experiment/robustness/failure evidence to Meta-Learning and bound next research choice.	Deterministic multi-cycle replay; no trading selection.
11 - Surface portfolio closure	Adjudicate/materialize each permitted surface separately; preserve excluded/constitutional families.	Every family authorized/deferred/excluded.
12 - Productive lineage closure	For each productive-relevant surface, prove candidate -> evidence -> governance/risk -> authorization -> governed parameter -> typed CURRENT consumer.	Complete auditable/replayable lineage.
13 - Promotion boundary	Implement/retain separate M10 governance boundary only under separate Owner authority.	No self-deploy; no optimizer direct write.
14 - Decision Attribution	Condition realized outcomes on existing MV2/DP refs only as evidence.	MV2/DP ownership unchanged.
15 - Final DoD	Adjudicate unified completion matrix against repo/runtime evidence.	UNKNOWN never closed by plausibility.

10. Mapping of the former roadmaps into the unified sequence

The former M0-M11 roadmap remains dependency-valid, but is now embedded beneath the Market-Intelligence sequence instead of being treated as a separate project.

Former item	Unified placement
M0 Self-Learning minimal closed loop	Phase 4/5 evidence baseline; retain CURRENT behavior.
M1 Learning Evidence Export + Optimization Input	Phase 8/9 inter-loop join.
M2 Optimization Universe Foundation + Registry	Phase 9 foundation.
M3 Optimizable Envelope + Resolver	Phase 9/11 surface authorization boundary.
M4 Experiment Plane / Search / Challenger / Robustness	Phase 9.
M5 Optimization Experiment Evidence Return	Phase 10.
M6 Meta-Learning Ingest + Meta-Evidence Export	Phase 10.
M7 Meta -> Optimization Research Feedback	Phase 10; experiment/search choice only.
M8 Deterministic multi-cycle replay	Phase 10 and final DoD.
M9 concrete optimization surfaces	Phase 11; one universe, multiple isolated surfaces.
M10 Governance/Risk promotion boundary	Phase 13; separate authority.
M11 optional bounded promotion automation	Outside default completion; requires its own ratification.

10.1 Surface ordering

F1/M9 remains the reference vertical proof. F2 Fee/Slippage may be considered only after explicit surface ratification. F5 must be decomposed/adjudicated. F3 requires an authority-boundary decision. Risk/Sizing remains excluded/constitutional. The unified blueprint does not infer authorization from historical code presence.

11. Unified Definition of Done

DoD family	Completion evidence
Data substrate	Public/private facts/state are canonicalized without duplicate transport truth; historical query/replay deterministic.
Market Intelligence	State/behavior/forecast/uncertainty/calibration/drift/coverage contracts have explicit provenance and abstention.
Temporal integrity	No lookahead; overlap control; finalization and horizon identity explicit.
Learning loop	Forecast/outcome and system evidence reach existing learning state/evidence with proven writer/reader/persistence semantics.
Optimization loop	Learning input -> experiment/search -> robustness/comparison -> evidence return is typed and replayable.
Meta loop	Experiment/failure evidence -> meta-evidence -> bounded next research choice is typed, deterministic and authority-free.
Failure memory	Rejected hypotheses and reasons are preserved and reused.
Surface governance	Every family is explicitly authorized, deferred or excluded; universe membership alone has no effect.
Productive relevance	Every productive-relevant surface identifies a concrete versioned CURRENT consumer.
Parameter lineage	Surface -> candidate -> evidence -> governance/risk -> explicit authorization -> governed parameter -> typed consumer is audit/replay complete.
Authority	No optimizer/meta path can write, authorize or deploy productive configuration directly; no MV2/DP mutation.
Trading authority	Cap 2.3 remains sole selection owner; MV2 + Double Play remain sole trading-decision authority.
External effect	Learning/Optimization/Market Intelligence create no direct order/wire path.

12. How to read status without creating confusion

The source documents were written at different repository baselines. Their architecture can be unified, but their historical CURRENT-status statements must not be copied forward as if they were all simultaneously current. Therefore this blueprint separates target architecture from execution status.

Rule: Target semantics are durable planning guidance. CURRENT implementation status is always re-established from origin/main before each bounded work package.

12.1 Durable target statements

• Public/private planes are governed fact/state substrates.

• Track D primarily learns market behavior and forecast reliability.

• Learning and Optimization reuse existing internal loops and typed evidence joins.

• Optimization is a single universe with isolated, explicitly authorized surfaces.

• Meta-Learning improves research choice, not trading choice.

• Productive parameter relevance requires full typed lineage and explicit authorization.

• Risk remains a hard governance boundary; Risk/Sizing is not an optimization surface under this blueprint.

• MV2 + Double Play remain sole trading-decision authority.

12.2 Status fields to re-prove before work

Baseline SHA, merged package status, D01/D02/D03 completion, current surface registry, F1/F2/F3/F5 state, current promotion/governance implementation, live pins, and any newly introduced productive consumer seams must be read from CURRENT rather than inferred from these source PDFs.

13. Unified Closure Position

The two former documents now resolve into one coherent end-state: trustworthy public/private runtime facts feed a Market-Intelligence layer that learns market behavior probabilistically; realized N_BARS outcomes close the forecast-learning loop; Learning evidence feeds a bounded Optimization Universe; Optimization evidence returns through Meta-Learning to improve future research; robust candidates remain PROPOSAL_ONLY until Governance and Risk explicitly authorize a concrete productive target; authorized values reach only typed CURRENT consumer seams; and Master V2 + Double Play continue to decide trades.

```
FACTS / STATE
-> MARKET INTELLIGENCE
-> REALIZED OUTCOMES
-> LEARNING EVIDENCE
-> OPTIMIZATION
<-> META-LEARNING
-> ROBUST CANDIDATE + EVIDENCE
-> PROPOSAL_ONLY
-> GOVERNANCE + RISK
-> EXPLICIT AUTHORIZATION
-> GOVERNED PRODUCTIVE PARAMETER
-> TYPED CURRENT CONSUMER
-> MASTER V2 + DOUBLE PLAY DECIDES
-> EXISTING PRETRADE / EXECUTION BOUNDARY
```

The intelligence of the system may grow substantially. The productive authority of Learning and Optimization does not.

Recommended next governed action

Use this unified blueprint as the single planning/navigation artifact (AUTHORITY=NONE). Before the next implementation slice, validate CURRENT origin/main and map the already completed D01/D02/D03 work onto Phase 0-15 above. Continue from the first unproven dependency rather than replaying completed work or following the historical status of either source PDF.

BLUEPRINT_AUTHORITY=NONE | FORECAST_IS_NOT_DECISION=true | CONFIDENCE_IS_NOT_AUTHORITY=true | LEARNING_IS_NOT_PROMOTION=true | OPTIMIZATION_IS_NOT_PROMOTION=true | NO_AUTOMATIC_PROMOTION=true | NO_SELF_DEPLOY=true

14. Market Context Fabric - Learning Information Expansion

This extension does not create a third intelligence stack and does not replace the two established evidence loops. It expands the information available to Market Intelligence and Learning while preserving the existing Public Data Plane, WebSocket strategy, N_BARS outcome semantics, Optimization Universe and authority boundaries.

PUBLIC_WS_STRATEGY_UNCHANGED=true | NO_SECOND_MARKET_TRUTH=true | CANONICAL_FACTS_FIRST=true | MARKET_CONTEXT_AUTHORITY=NONE | MV2_DP_UNCHANGED=true

14.1 Canonical information flow

CANONICAL MARKET FACTS -> MARKET CONTEXT(t) -> FORECAST(t,N) -> REALIZED BEHAVIOR(t+N) -> CALIBRATION / CONDITIONED LEARNING EVIDENCE -> OPTIMIZATION INPUT

Market Context is a typed representation layer inside the MI/Learning information path. It may derive context from canonical facts and separately governed fact adapters, but it must not become a competing market truth, a trading signal owner, or a productive decision owner.

14.2 Context families

Family	Target information	Rule
PRICE_STATE	Returns, momentum/range, extrema, multi-horizon price structure.	Derive from canonical price facts; no duplicate price SSOT.
FLOW_STATE	Signed trade volume, buy/sell imbalance, trade intensity, size distribution, price response to flow.	Derive from canonical trades where semantics support it.
LIQUIDITY / MICROSTRUCTURE	Spread state/dynamics plus admissible CURRENT microstructure evidence.	Reuse existing microstructure first; distinguish proxy from true L2.
DERIVATIVES_STATE	Mark/index/last basis, open interest level/change, funding state/extremeness.	Add only typed orthogonal facts not already represented.
VOLATILITY_STATE	Short/medium/long realized vol, ratios, expansion/compression, vol-of-vol.	Derived, horizon-explicit and point-in-time safe.
CROSS_MARKET_STATE	Bounded anchors such as BTC/ETH vs selected future: relative return/vol, correlation/beta, common/idiosyncratic move.	Context only; never productive selection.

14.3 Anti-feature-soup law

NO_FEATURE_ADMISSION_BY_AVAILABILITY=true. Every context family must be admitted, deferred or rejected through incremental evidence. Rejected and redundant families remain first-class research/meta evidence.

15. Canonical Market Context and Realized Behavior Contracts

The implementation target is a compositional context identity, not a monolithic feature vector with hidden ownership. Context references preserve source identity, observation time, provenance, quality and feature-version lineage.

15.1 MARKET_CONTEXT_V1 target shape

MARKET_CONTEXT_V1 = { context_id, observed_at, instrument_ref, information_set_ref, price_state_ref, flow_state_ref, liquidity_microstructure_state_ref, derivatives_state_ref, volatility_state_ref, cross_market_state_ref, quality_state_ref, provenance_refs, feature_versions }

Missing context families remain explicit. No silent zero-fill, forward-fill or inferred availability is permitted where the source contract does not authorize it.

15.2 Realized behavior target

Behavior family	Examples	Evidence role
Forward behavior	Forward return distribution; continuation/reversal/range.	Condition forecast and representation quality.
Excursion	MFE / MAE over explicit N_BARS horizon.	Measure path quality, not only terminal return.
Volatility / tails	Realized volatility, tail/shock behavior.	Measure uncertainty and adverse-state behavior.
Transition	State/regime transition labels where supported.	Measure conditional stability and transition risk.

15.3 Normative temporal join

MARKET_CONTEXT(t) -> FORECAST(t,N) -> REALIZED_BEHAVIOR(t+N) -> ERROR / CALIBRATION -> CONDITIONED_LEARNING_EVIDENCE

The join must preserve point-in-time information sets, finalization, explicit horizon identity, overlap control and lookahead prevention. N_BARS remains the normative realized-outcome backbone; this extension enriches conditioning information rather than replacing the outcome mechanism.

15.4 Decision attribution remains subordinate

ATTRIBUTION_EVIDENCE_V1 = { market_context_ref(t), selected_future_ref, mv2_dp_decision_ref(t), side_state_ref, configuration_ref, realized_behavior_ref(t+N), quality_refs }

Attribution measures under which contexts existing MV2/Double Play decisions were robust, weakly calibrated or insufficiently supported. It is evidence only and creates no runtime gate or decision mutation.

16. Strongest Closed Intelligence Cycle - Three Learning Functions

The target architecture preserves Loop A and the established Optimization/Meta loop, then adds one typed Meta-to-Learning feedback seam. Meta-Learning does not broadcast undifferentiated evidence. It classifies evidence by semantic owner and returns it only to the stack that is allowed to use it.

MARKET -> LEARNING -> OPTIMIZATION -> META-LEARNING -> { OPTIMIZATION | LEARNING } -> NEXT CYCLE

16.1 Loop A - Market / Outcome Learning (preserved)

PRODUCTIVE OBSERVATION / DECISION + MARKET_CONTEXT(t) -> REALIZED OUTCOME / SYSTEM EVIDENCE -> SELF-LEARNING -> LEARNING EVIDENCE -> OPTIMIZATION INPUT

Purpose: learn market behavior, forecast reliability, conditioned outcomes, calibration, coverage and drift.

16.2 Loop B - Research / Optimization Meta Loop (preserved)

OPTIMIZATION EXPERIMENT -> CHALLENGER / OOS / ROBUSTNESS / FAILURE EVIDENCE -> META-LEARNING -> META_RESEARCH_EVIDENCE -> BETTER NEXT OPTIMIZATION RESEARCH / SEARCH CHOICE

Purpose: learn which search families, surfaces, experiment structures and research choices produce robust or repeatedly rejected evidence.

16.3 Loop C - Representation Feedback Loop (new)

LEARNING REPRESENTATION -> OPTIMIZATION EVIDENCE -> META-LEARNING -> META_LEARNING_EVIDENCE -> LEARNING RESEARCH / ADAPTATION INPUT -> NEW LEARNING EVIDENCE

Purpose: return evidence about feature usefulness, redundancy, context-family value, regime conditionality, coverage weakness, representation drift and repeatedly failed representation hypotheses to the Learning research context.

16.4 Core law

META_FEEDBACK_IS_EVIDENCE_NOT_AUTHORITY=true | NO_SELF_MODIFYING_LEARNING=true | NO_DIRECT_FEATURE_DROP=true | NO_DIRECT_PARAMETER_MUTATION=true | NO_TRADING_GATE_FROM_META=true

17. Typed Meta-Evidence Routing - No Broadcast

The strongest closed cycle requires semantic routing. A Meta-Learning result is not sent to both stacks merely because both could consume it. Each record declares its evidence class, intended owner, provenance and permitted use. Unknown or mixed classifications fail closed.

17.1 META_EVIDENCE_V1 routing classes

Evidence class	Canonical consumer	Examples	Permitted effect
RESEARCH_CHOICE	Optimization	Search-family performance; challenger effectiveness; parameter-space efficiency; experiment failure patterns; surface prioritization.	Bounded next experiment/search prioritization only.
LEARNING_REPRESENTATION	Learning	Feature usefulness/redundancy; context-family usefulness; regime conditionality; coverage weakness; representation drift; failed representation hypotheses.	Learning research/adaptation input only; no direct mutation.
UNKNOWN / MIXED	None automatically	Ambiguous ownership, cross-cutting evidence, unresolved semantics.	Persist and adjudicate; no automatic feedback.

17.2 Routing law

RESEARCH_CHOICE -> OPTIMIZATION ONLY | LEARNING_REPRESENTATION -> LEARNING ONLY | UNKNOWN/MIXED -> NO AUTOMATIC FEEDBACK

17.3 Learning-side consumption law

Meta-to-Learning feedback may influence only bounded research/adaptation choices that are explicitly represented in the Learning contract. It may not silently change productive features, model weights, thresholds, trading configuration, selection, risk, sizing or execution behavior.

17.4 Optimization-side consumption law

Meta-to-Optimization feedback may prioritize the next bounded experiment/search choice. It may not promote candidates, write productive configuration, authorize a surface, select a productive future, mutate MV2/Double Play or create an external effect.

17.5 Deterministic replay

A complete cycle must replay: source evidence -> meta ingest -> classification -> routed meta evidence -> consumer research choice -> next evidence generation, with stable identity and provenance.

18. Incremental Information Program and Optimization Combination

The expanded Learning information path and Optimization Universe are combined through the existing typed Learning Evidence -> Optimization Input seam. Context families are not admitted because they exist; their incremental value is tested against a preceding baseline.

18.1 Incremental ladder

Stage	Information set	Research question
B0	Existing canonical Price + Flow baseline.	What is achievable from CURRENT canonical facts?
B1	B0 + admissible CURRENT Microstructure.	Does existing microstructure add stable incremental information?
B2	B1 + Derivatives State.	Do basis/OI/funding add orthogonal OOS information?
B3	B2 + Cross-Market State.	Does bounded anchor context improve conditional behavior/calibration?
B4	B3 + Volatility Structure.	Does multi-horizon volatility/liquidity structure add robust value?
B5	B4 + true L2 Depth, optional.	Only if incremental OOS evidence justifies added runtime/data complexity.

18.2 Required evaluation dimensions

Incremental information versus prior baseline; walk-forward/OOS stability; calibration; coverage/abstention; regime/state stability; redundancy; drift sensitivity; temporal integrity; and operational cost/complexity. Failure and rejection remain evidence for Meta-Learning.

18.3 Microstructure rule

MICROSTRUCTURE_REUSE_FIRST=true. Existing admissible microstructure is inventoried and reused before any new producer is created. Proxy microstructure and true tick/L2 microstructure must never be silently conflated.

18.4 Cross-market rule

CROSS_MARKET_IS_CONTEXT_ONLY=true. Anchor instruments may inform context/evidence but cannot rerank, reselect or create multi-future productive trading authority.

18.5 Productive boundary

Even a robust context family or optimized representation remains evidence/research output. Any productive parameter relevance still follows the existing candidate -> evidence -> Governance/Risk -> explicit authorization -> governed parameter -> typed CURRENT consumer lineage.

19. Canonical Implementation Sequence - Closed-Cycle Extension

This extension begins only after CURRENT origin/main is revalidated. It does not replay already proven work. Each phase is a bounded work package and stops at the first real authority, semantic or technical blocker.

Phase	Action	Exit / stop condition
16 - Context Census	Inventory CURRENT Price/Flow/Liquidity/Microstructure/Derivatives/Volatility/Cross-Market producers, contracts, stores, adapters and consumers. Classify raw/derived/proxy and duplication.	No invented owner; missing orthogonal facts explicit.
17 - Context Contracts	Ratify MARKET_CONTEXT_V1, information-set identity, provenance, missing semantics, feature versions and PIT rules.	No second market truth; N_BARS semantics preserved.
18 - Existing-Fact Context	Materialize Price/Flow/Liquidity/Volatility from canonical facts and reuse admissible microstructure.	Deterministic/replayable context from CURRENT facts.
19 - Orthogonal Context	Add only adjudicated missing Derivatives and bounded Cross-Market fact/adapters.	Typed provenance; no WS strategy redesign; context only.
20 - Behavior Join	Join context(t) to N_BARS realized behavior incl. forward return, MFE/MAE, realized vol and typed labels.	PIT-safe, horizon-explicit, overlap controlled.
21 - Learning Integration	Feed conditioned context/outcome/calibration evidence through existing Learning state/evidence path.	Loop A closed; no trading/promotion authority.
22 - Incremental Research	Run B0-B5 through Optimization with challenger/OOS/robustness/failure evidence.	Each family admitted/deferred/rejected by evidence.
23 - Meta Dual Routing	Extend Meta-Learning with typed RESEARCH_CHOICE and LEARNING_REPRESENTATION evidence routing.	Optimization and Learning consumers replay deterministically; UNKNOWN/MIXED fail closed.
24 - Representation Feedback	Consume LEARNING_REPRESENTATION meta evidence as bounded Learning research/adaptation input.	Loop C closed; no self-modification/direct feature mutation.
25 - DP Attribution	Compose existing MV2/DP refs with context + realized behavior as evidence only.	Decision ownership unchanged.
26 - Final Closed-Cycle DoD	Adjudicate complete three-loop topology, routing, authority and replay against CURRENT evidence.	UNKNOWN never closed by plausibility.

19.1 Closed-cycle Definition of Done

LOOP_A_PROVEN + LOOP_B_PROVEN + LOOP_C_PROVEN | META_ROUTING_TYPED | META_TO_LEARNING_EVIDENCE_ONLY | MARKET_CONTEXT_TYPED | INCREMENTAL_OOS_EVIDENCE | NO_SECOND_MARKET_TRUTH | NO_AUTHORITY_EXPANSION

19.2 Target pins

PUBLIC_WS_STRATEGY_UNCHANGED=true | MARKET_CONTEXT_AUTHORITY=NONE | META_EVIDENCE_AUTHORITY=NONE | OPTIMIZATION_PROMOTION_AUTHORITY=NONE | LEARNING_IS_NOT_PROMOTION=true | MV2_DP_UNCHANGED=true | NO_AUTOMATIC_PROMOTION=true | NO_SELF_DEPLOY=true

Recommended next governed action

Revalidate CURRENT origin/main, then execute Phase 16 as a read-only Context Census. Preserve the existing WebSocket strategy and established loop topology; the first implementation mutation begins only after the census proves the exact CURRENT owners, gaps, derivability and duplicate surfaces.